NIHAL SINGH

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EXPERIENCE

Amazon, Industrial Robotics Group — Bellevue, WA
Senior Software Development Engineer / Tech Lead — Apr 2026 – Present

- Built **Kubernetes/Metaflow training infrastructure** to serve 30+ scientists on ~1,800 GPUs (~225 nodes) across AWS and Lambda Labs; building Kueue priority queuing and fair-share scheduling across teams.
- Built a **robotics data platform** for ~150 TB / ~1,000 hours of internal data, managing the entire lifecycle from collection and processing to dataset creation and model training, with lineage and quality gates.
- Cut **pre-training from ~30 to ~9 days** on the same 128 GPUs with FSDP/DDP, PyTorch compilation, and NCCL profiling; cut RL training from ~6 hours to ~20 minutes by tuning parallel environments, batches, and rollouts.
- Built **annotation workflows** for language, segmentation masks, and task labels: ~150 hours of ground truth from an internal workforce; ~10,000 hours (incl. open datasets) using frontier models as VLMs.
- Automated **simulation benchmarks and model selection**, prioritizing top policies for real-robot testing.
- Built **real-to-sim data augmentation** to expand real demos into diverse simulated training trajectories.

Amazon, Sustainability Technology — Seattle, WA
Software Development Engineer II → Senior Software Development Engineer — Oct 2022 – Apr 2026

- Led an 18-person team building Amazon's **carbon-accounting platform** for regulatory reporting; designed git-style versioning of TB-scale data with lineage across ~200 annual methodology and emission-factor revisions.
- Cut **ETL runtime from 12+ hours to <3 hours** by consolidating Spark services into a Polars monolith, eliminating shuffles and short-circuiting unchanged graph steps; built custom Iceberg CDC parallelized by time partition.
- Architected a **zero-downtime Iceberg migration**, reducing storage from 600 TB to <20 TB and critical query latency from 30 minutes to 5 seconds; retired legacy Hive versioned storage, saving $1.26M/year.
- Led six engineers to **automate utility invoice processing**: PDF transcription, reconciliation, calendarization, gap estimation, and REC allocation; cut data lag from 90 to 30 days for faster facility corrective action.

Amazon, IN Personalization Seattle, WA **Software Engineer I, Intern** — 2021 – 2022 (Intern, 2020)

- Built **personalized deal recommendations** using customer-proficiency scores and browsing behavior.
- Deployed **ML and heuristic models** for lighter Amazon.in experiences on slow connections.

Mercari Tokyo, Japan **Android Engineering Intern** — 2018

- Simplified item creation with **Android listing templates** for Mercari, Japan's largest C2C marketplace.

EDUCATION

Georgia Institute of Technology, MS in Computer Science, GPA 4.0/4.0 — 2019 – 2020
Indian Institute of Technology Bombay, B. Tech in Civil Engineering, CS minor, CGPA: 8.74/10.00 — 2015 – 2019

TECHNICAL SKILLS

Python, Java, Kubernetes, Spark, Polars, Iceberg, AWS (ECS, EKS, Step Functions, Lambda, OpenSearch, Batch)

MISCELLANEOUS

- **Google Summer of Code:** Terasology contributor (2017); GSoC and Google Code-In mentor (2016–2019).
- **Coding club:** Led a team of 22 at IIT Bombay, organizing talks, workshops, and competitions.
- **Hackathon awards:** Smart India, Microsoft Code.Fun.Do, Yahoo! Japan, Ubisoft and Lenovo Game Jams.
- **Interests:** Meditation, hiking, running, reading science fiction, and progressive rock.